

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)	(i)		Suitable scales, e.g. 0, 20, 40, 60 on the x -axis and 0, 2, 4, 6 on the y -axis (1) 5 points correctly plotted $\pm < 1$ small square (2) 4 points correctly plotted $\pm < 1$ small square (1) 3 points or less correctly plotted $\pm < 1$ small square (0) Smooth curve of best fit $\pm < 1$ small square (1)	1	2 1		4	4	4
		(ii)		As temperature increases resistance decreases (1) at a decreasing rate (1)		2		2		2
	(b)	(i)		Resistance correct from their graph (1) e.g. 8 [k Ω] $\frac{1}{R} = \frac{1}{8} + \frac{1}{5}$ (1) substitution $R = 3.08$ [k Ω] (1) $V = IR$ $12 = I \times 3.08 \times 10^3$ ecf on R (1) substitution $I = \frac{12}{3.08 \times 10^3} = 0.0039$ [A] (1) ecf Answer of 3.9×10^n where n is not -3 award 4 marks Alternative [for marks 2 – 5] $I = \frac{V}{R} = \frac{12}{5} [= 2.4 \text{ mA}]$ (1) substitution $I = \frac{V}{R} = \frac{12}{8} [= 1.5 \text{ mA}]$ (1) substitution Total $I = 2.4 + 1.5 = 3.9$ [mA] (1) ecf addition of currents $I = 0.0039$ [A] (1)	1 1	1 1		5	5	5